

## Suggested FYP topics for 2026-27

by

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| Project code :             | CHZ1                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Project title :            | Power network topology optimization                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| No. of students :          | 1~2                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Description of project :   | Solve optimal topology and power flow (OTPF) problem that jointly optimizes on/off status of power lines and power generations, loads, power flows, for reliable and efficient operation with high percentage of renewable generation. Develop an OTPF solution algorithm with improved computation efficiency and solution quality. Main tasks include literature survey and study, computer simulation programming (e.g., in MATLAB), and algorithm development and testing. |
| Industrial collaboration : | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| Project code :             | CHZ2                                                                                                                                                                                                                                                                                                                                                                                                     |
| Project title :            | Improving AC power flow solution                                                                                                                                                                                                                                                                                                                                                                         |
| No. of students :          | 1~2                                                                                                                                                                                                                                                                                                                                                                                                      |
| Description of project :   | Solve the fundamental AC power flow problem, where given some variables in nonlinear AC power flow equations, solve for other variables. Improve the convergence rate of the existing Backward-Forward Sweep (BFS) algorithm for solving power flow equations. Main tasks include literature survey and study, computer simulation programming (e.g., in MATLAB), and algorithm development and testing. |
| Industrial collaboration : | No                                                                                                                                                                                                                                                                                                                                                                                                       |

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| Project code :             | CHZ3                                                                                                                                                                                                                                                                                                                                                                       |
| Project title :            | Self-proposed topic                                                                                                                                                                                                                                                                                                                                                        |
| No. of students :          | 1~2                                                                                                                                                                                                                                                                                                                                                                        |
| Description of project :   | Any topic that applies mathematics to solve a practical engineering problem. Preference for control and dynamical systems, stochastic processes, optimization, machine learning, game theory, and networked systems. Please email <a href="mailto:chzhao@ie.cuhk.edu.hk">chzhao@ie.cuhk.edu.hk</a> a half-page project proposal for discussion and approval of your topic. |
| Industrial collaboration : | No                                                                                                                                                                                                                                                                                                                                                                         |